



Report

Robot Mapping with Rplidar

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1. Pre-requisites

- Ubuntu 24.04 Noble can run on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This is NOT a simulation. Gazebo is no longer used. The hardware is required.
- VSCode and its extension PlatformIO (PIO) should be installed.
- First, connect the ESP32 to the computer to test the communication between ROS2 and the hardware. Later, it would be mandatory to connect the RPI to the ESP32, and ssh into the RPI.

2. Tasks

2.1. With the Rplidar A1M8 connected on the computer

At first it is necessary to verify that the Rplidar A1M8 is working. Please install the required packages.

Tasks	Code
Plug the RPLidar to the computer	
Install the package for the lidar hardware (rplidar A1M8)	<code>sudo apt install ros-jazzy-rplidar-ros</code>
Run the lidar (hardware) Be aware about the name of the serial port: <code>/dev/ttyUSB0</code>	<code>ros2 run rplidar_ros rplidar_composition --ros-args -p serial_port:=/dev/ttyUSB0 -p frame_id:=lidar_link -p angle_compensate:=true -p scan_mode:=Standard</code>
Run rviz2	<code>ros2 run rviz2 rviz2</code>

You should be able to visualize the cloud point from the lidar in rviz.

2.2. With the Rplidar A1M8 connected on the RPI

So far, the computer is connected to the Lidar. A more common setup is as shown in Figure 1.

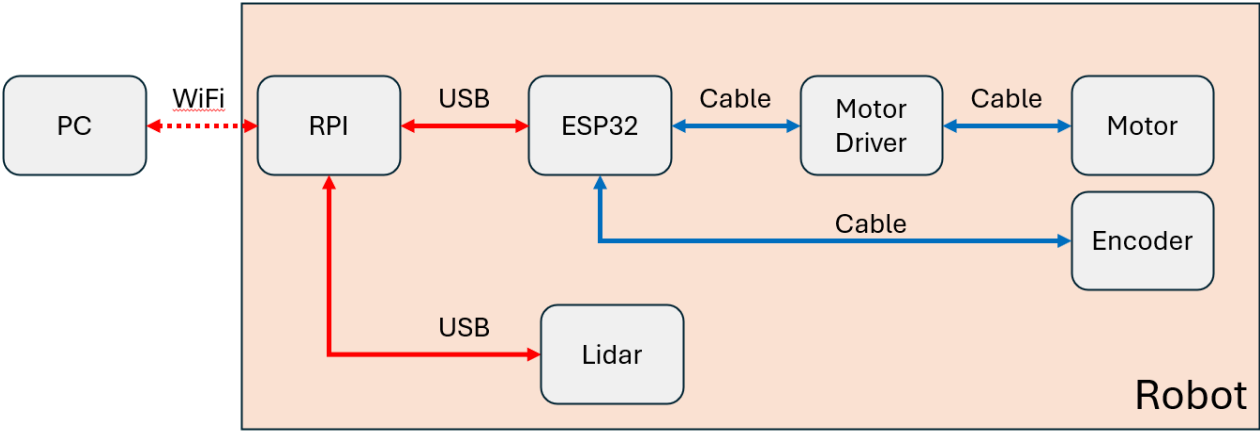


Figure 1: Robot configuration and connection

Tasks	Code
Start SSH (make sure the RPI and PC are connected on the same WiFi)	<code>ssh alpha-pi@alpha-pi.local</code>
On the RPI Install the package for the lidar hardware (rplidar A1M8)	<code>sudo apt install ros-jazzy-rplidar-ros</code>
On the RPI Run the lidar (hardware)	<code>ros2 run rplidar_ros rplidar_composition --ros-args -p serial_port:=/dev/ttyUSB0 -p frame_id:=lidar_link -p angle_compensate:=true -p scan_mode:=Standard</code>

You should be able to run the lidar from the RPI via SSH from the computer.

2.3. Consideration regarding serial port name with Ubuntu

Ubuntu dynamically changes the USB port name: if you connect your esp32 and your lidar, the usb name might change. Thus, it is advised to develop a user development rule (udev), on your RPI and on your PC, so that the name does not change when you plug your peripheral.

Tasks	Code
Use the command lsusb to know the information of your serial device	<i>lsusb</i> E.g.: esp32 usb information Bus 003 Device 020: ID 1a86:7523 QinHeng Electronics CH340 serial converter
Launch nano (text editing software in terminal):	<code>sudo nano /etc/udev/rules.d/99-serial-aliases.rules</code>
Copy the following lines:	<code>SUBSYSTEM=="tty",ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523", SYMLINK+="esp32"</code> <code>SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60", SYMLINK+="rplidar"</code>
Save:	Ctrl + O, Press Enter
Close:	Ctrl + X
in a terminal, launch the next CLI:	<code>sudo udevadm control --reload-rules</code> <code>sudo udevadm trigger</code>
Verify by launching: You should see:	<code>ls /dev/*</code> <code>/dev/esp32</code> <code>/dev/rplidar</code>

3. Explanation

There are few modifications that have been performed in the packages:

- use_sim_time parameter has been set to false in all packages
- The urdf is calling the hardware interface instead of the gazebo plugin.
- The integration of the Lidar hardware is located in rmitbot_mapping.

4. Packages, nodes, and topics

Packages	Nodes	Description
rmitbot_bringup		Launch the other packages and nodes
rmitbot_description	display.launch.py: rviz, rsp gazebo.launch.py: gazebo	Launch the urdf and static tf
rmitbot_controller	Controller manager: jsb spawner, controller spawner	Launch input/output of the robot
rmitbot_localization	ekf	Launch the sensor fusion between wheel encoder and imu
rmitbot_mapping	Slam rplidar	Launch the lidar and slam
rmitbot_navigation	nav2	Launch the navigation
rmitbot_vision	apriltag	Include the plugin for camera. Launch the apriltag detection
rmitbot_firmware	Arduino firmware Hardware interface	The arduino firmware is the code flashed on the esp32. The hardware interface is the node launched with ROS2.

5. Future works

The Rplidar should be launched from your RPI. The slamtoolbox should be launched from your computer. Can you move your robot and start to do the mapping of the area around the robot?